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SEMINAR AND PROJECT INITIATION COURSE

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Topic : ML – Based ATS Resume Optimizer

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Abstract

The **ML-Based ATS Resume Analyzer** and **Optimizer** is a proposed system that analyzes a resume against a selected job description. Text is extracted directly from digital resumes; **Optical Character Recognition (OCR)** is applied only when a resume is scanned or image-based. **Natural Language Processing (NLP)** identifies skills, education, experience, projects, and certifications in the resume, and required skills and criteria in the job description. **A Machine Learning (ML)** matching module estimates compatibility and finds matched, missing, and unverified requirements. A **Large Language Model (LLM)** then uses these results and the source evidence to explain skill gaps and suggest resume improvements, which the user reviews before use. The system is at the proposal stage; implementation and testing are future work.

1. Introduction

In today's competitive job market, candidates apply for many jobs, but they often find it difficult to prepare a resume that matches specific job requirements. Employers also use Applicant Tracking Systems (ATS) to screen and filter a large number of resumes. We chose the ML-Based Resume Optimizer topic because many candidates struggle to identify missing skills, improve resume content, and understand how well their resumes match a particular job description.

The proposed **ML-Based Resume Optimizer** aims to help candidates improve their resumes through the following stages:

- **OCR** : Extracts text from scanned or image-based resumes.
- **NLP** : Identifies important information such as skills, education, and work experience.
- **ML** : Calculates resume-job matching scores and identifies skill gaps.
- **Resume Optimization** : Provides suggestions to improve resume content and highlight relevant skills.

This project can help candidates prepare better-targeted resumes, identify areas for improvement, and make the job application process more effective. The matching score provides guidance only and does not guarantee ATS selection or employment.

2. Literature Survey

Sr. No.	Category	Paper Title
1	**OCR (STAGE 1)**	Resume Information Extraction via Post-OCR Text Processing
2	**NLP (STAGE 2)**	Technical Skills Information Extraction from Resumes Using Advanced Natural Language Processing Model with Transfer Learning
3	**NLP (STAGE 2)**	Resume Screening with Natural Language Processing (NLP)
4	**NLP (STAGE 2)**	Smart Hiring: An NLP and Ensemble-Based Approach to AI-Driven Resume Screening
5	**ML (STAGE 3)**	Resume Screening Using Machine Learning
6	**ML (STAGE 3)**	Automated Resume Screening Using Machine Learning: An Ensemble-Based Approach for Efficient Candidate Selection
7	**ML (STAGE 3)**	Design and Development of Machine Learning Based Resume Ranking System
8	**LLM (STAGE 4)**	Optimizing Resume Design for ATS Compatibility: A Large Language Model Approach
9	**LLM (STAGE 4)**	Resume Analyzer Using LLM
10	**LLM (STAGE 4)**	An LLM-Based Resume Intelligence System for Recruiters and Job Seekers

3. Information to Seminar topic

The proposed system follows four stages: OCR → NLP → Machine Learning → LLM. Each stage performs a specific task to analyze resumes and compare them with job requirements.

1. Optical Character Recognition (OCR) : OCR **extracts text** from scanned or image-based resumes. For digital PDF and DOCX files containing readable text, direct text extraction is used instead of OCR. This avoids unnecessary processing and supports different resume formats.

2. Natural Language Processing (NLP) : NLP **cleans** and **normalizes** the extracted text. It identifies relevant information from two sources: the **resume**, including **skills**, **education**, experience, projects, and certifications; and the job description, including required skills, qualifications, and experience. This converts **unstructured text** into information suitable for **comparison**.

3. Machine Learning (ML) : The ML stage compares the structured resume information with the job requirements. The proposed outputs include an estimated **compatibility score, matched requirements, missing requirements, unverified requirements,** and **skill-gap analysis**. These results help candidates understand how closely their resumes align with a **particular job.**

4. Large Language Model (LLM) : The LLM uses the **matching results** and supporting resume evidence to explain **skill gaps** and suggest clearer wording for existing qualifications and experience. It must not invent skills or qualifications. Suggestions should be checked against the original resume and reviewed by the user before use.

3.1 Overall System Workflow

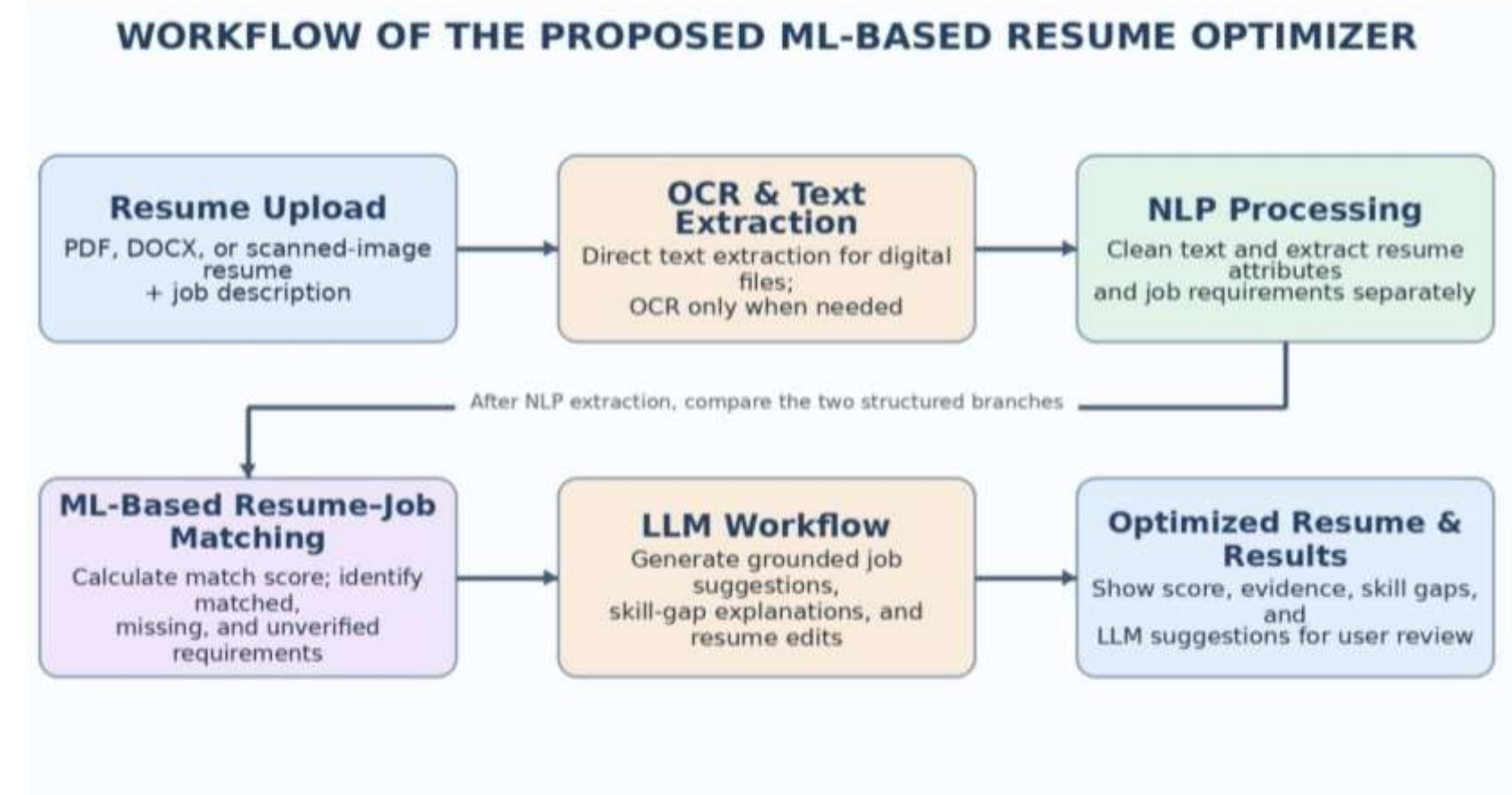


Fig. 3.2: Workflow of the Proposed ML – Based Resume Optimizer

3.2 Proposed System Architecture

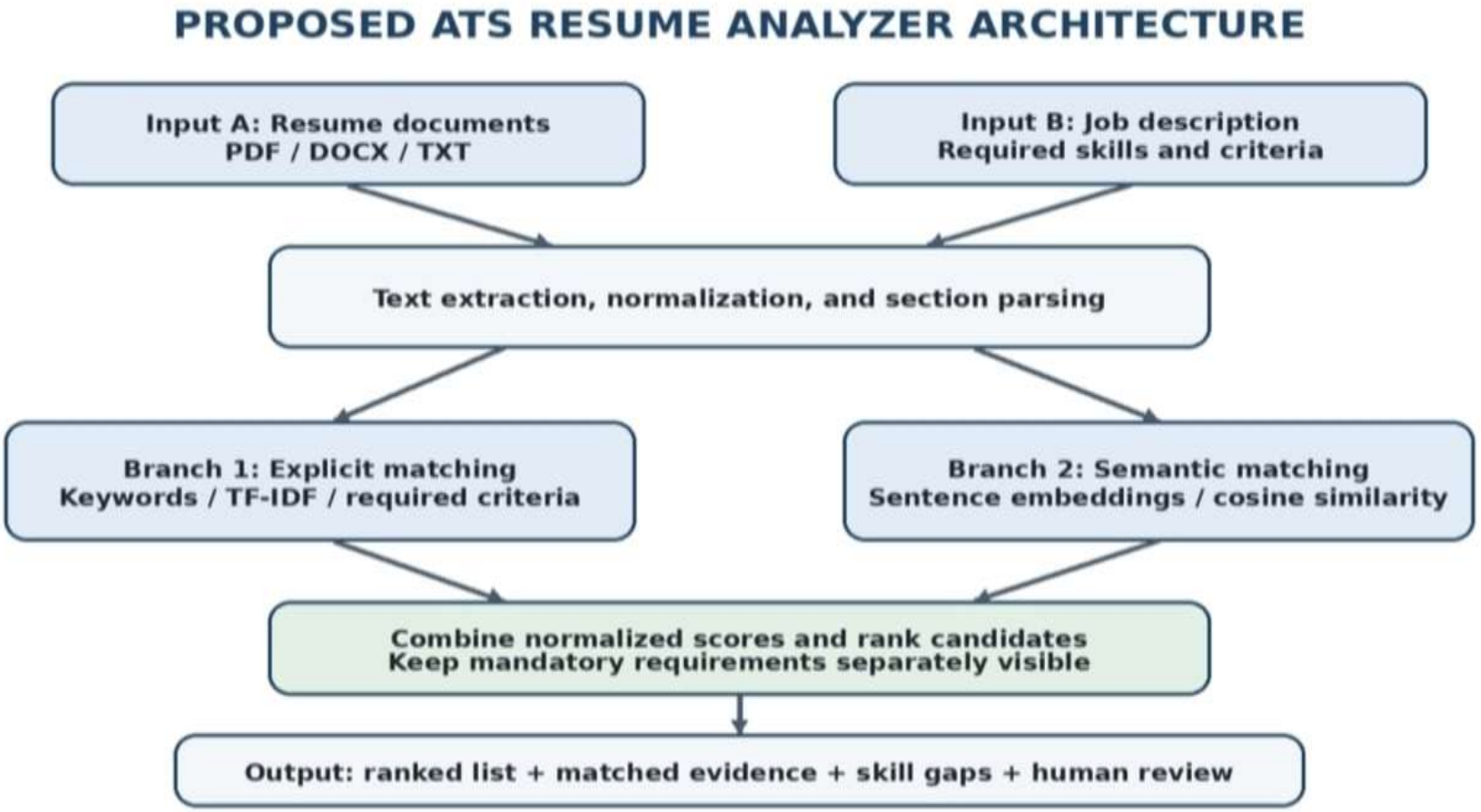


Fig. 3.1: Proposed ATS Resume Analyzer Architecture

3.3 Detailed Working of the System

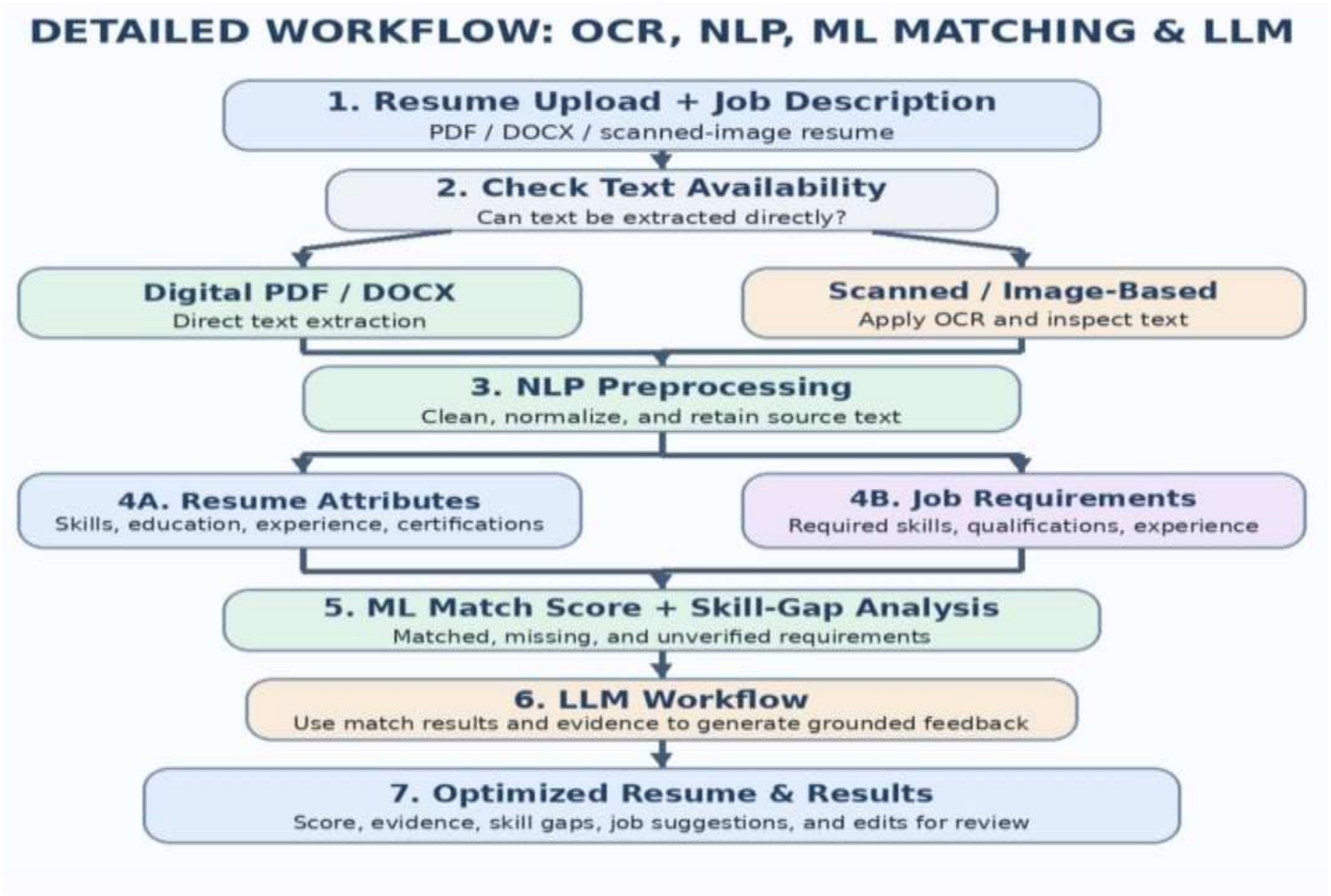


Fig. 3.3: Detailed Working of the System

3.4 OCR Based Resume Workflow

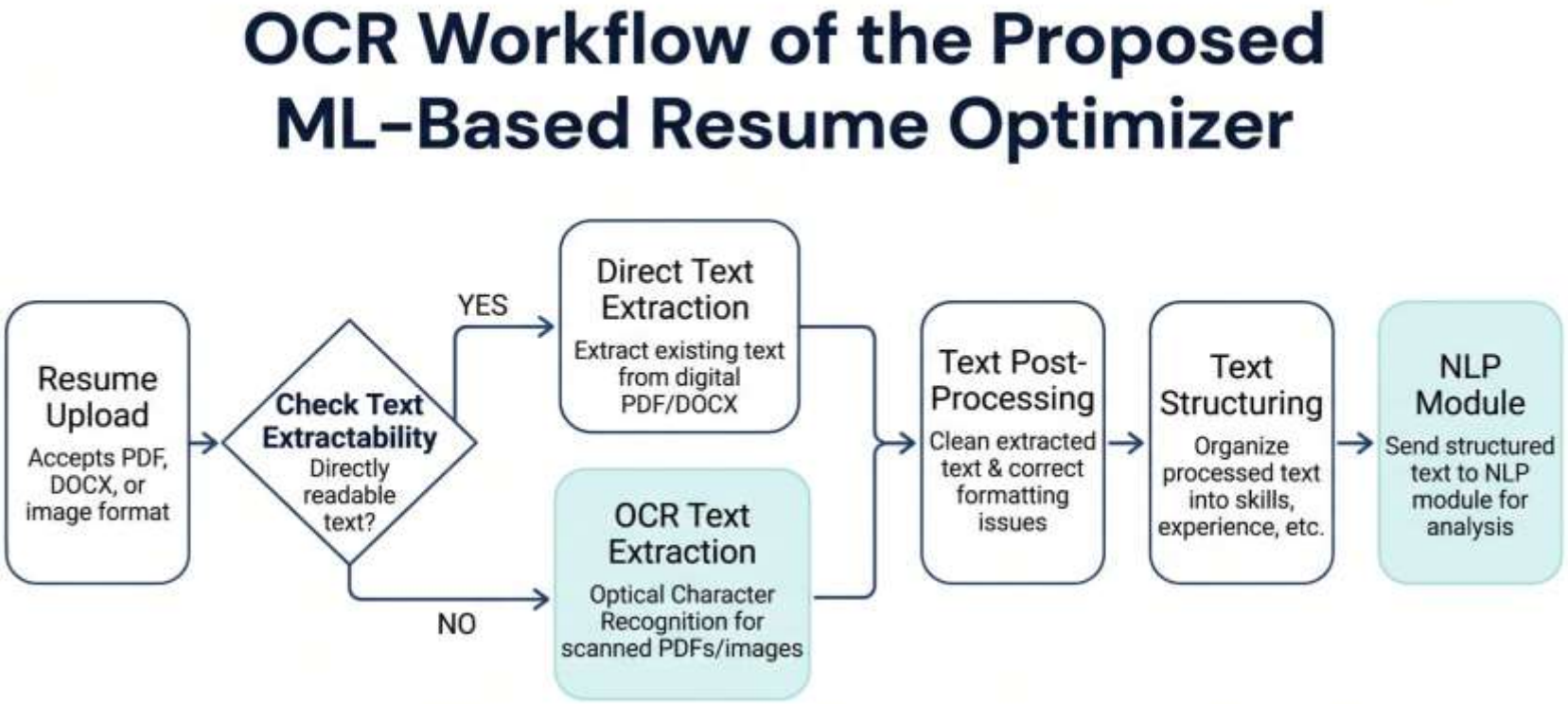


Fig. 3.4: OCR Workflow of the Proposed ML – Based Resume Optimizer

3.5 NLP Module Workflow

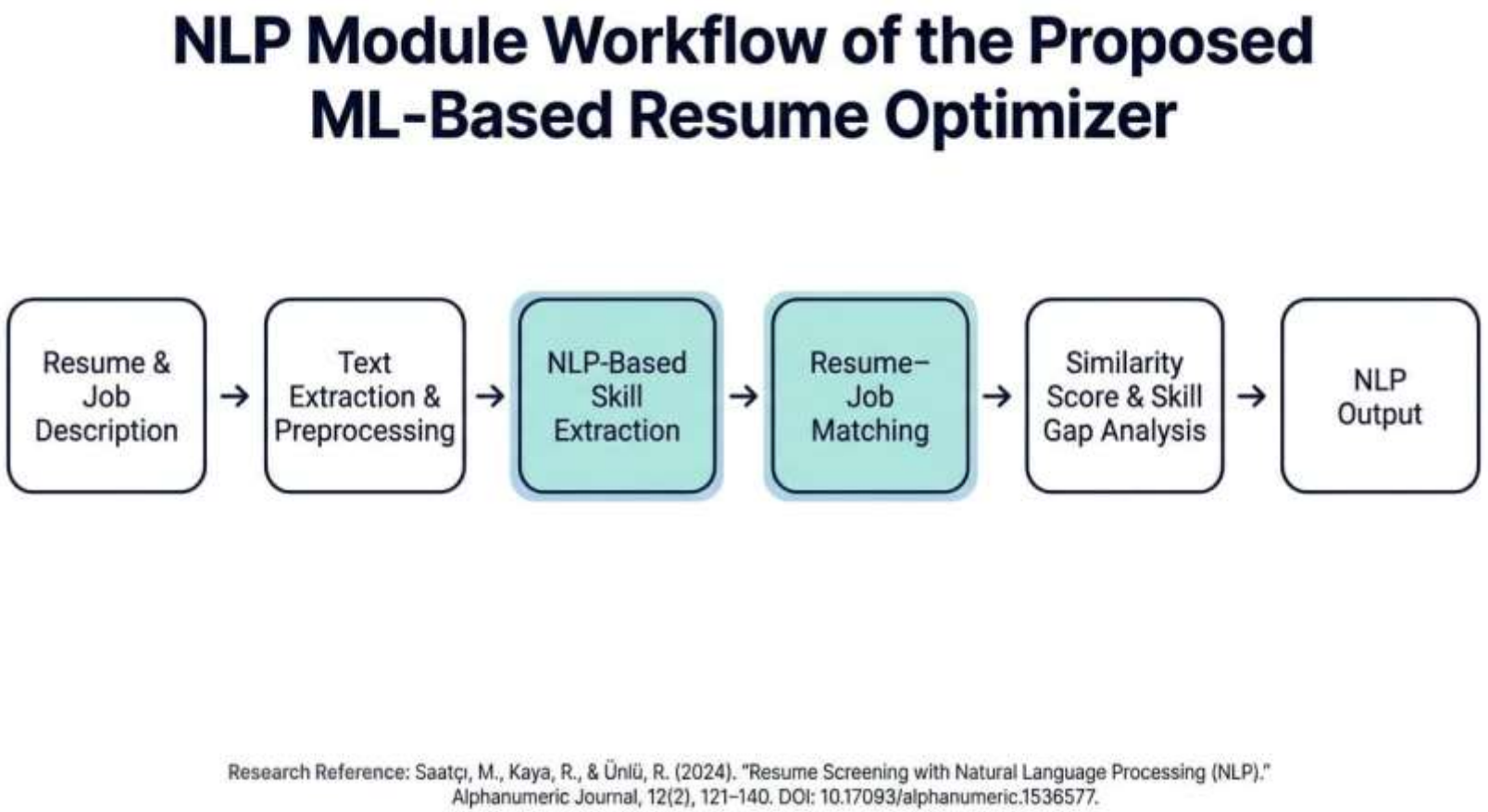


Fig. 3.5: NLP Workflow : Resume & Job Information Extraction

3.6 ML System Workflow

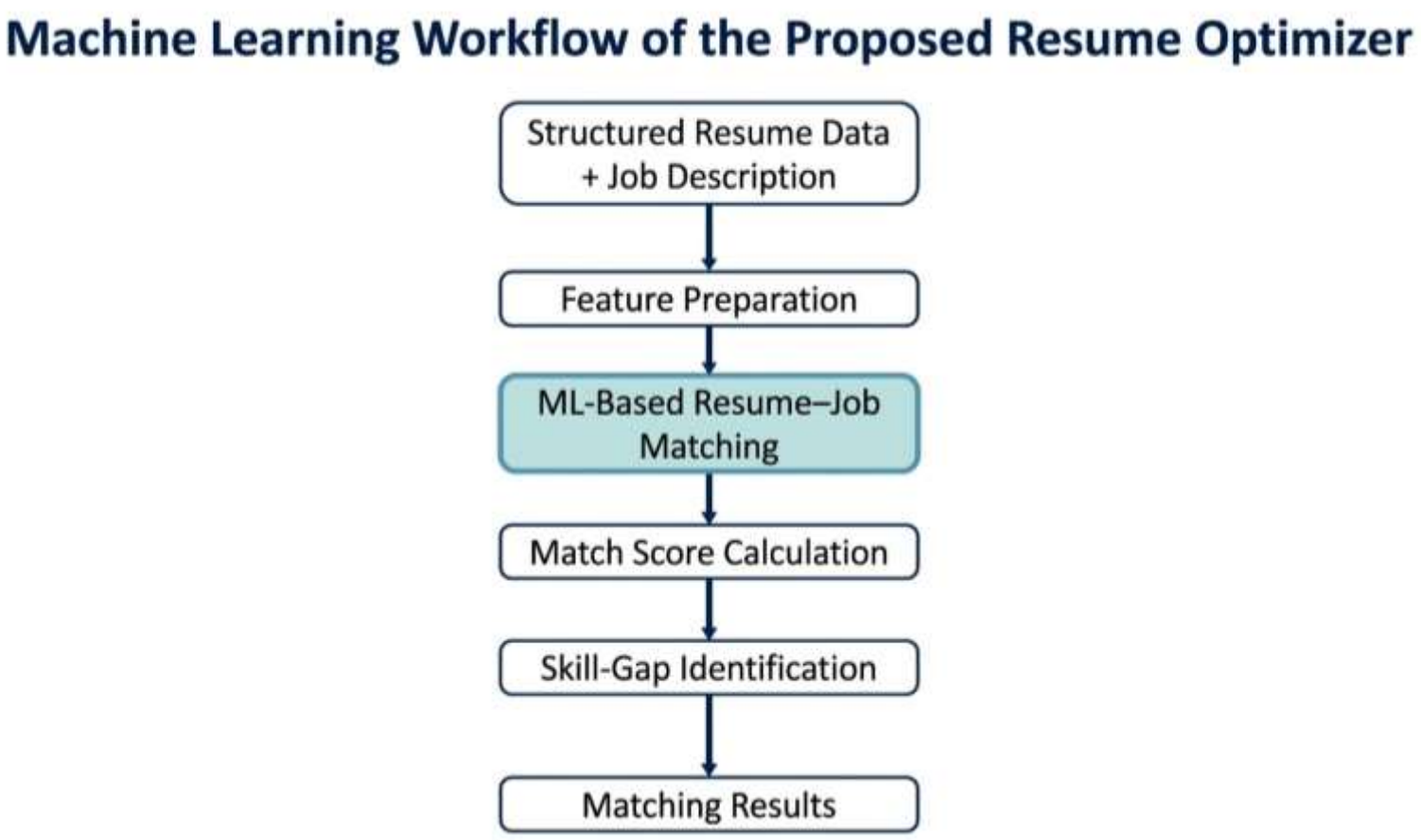


Fig. 3.6: Original Machine Learning Workflow

3.7 LLM Module Workflow

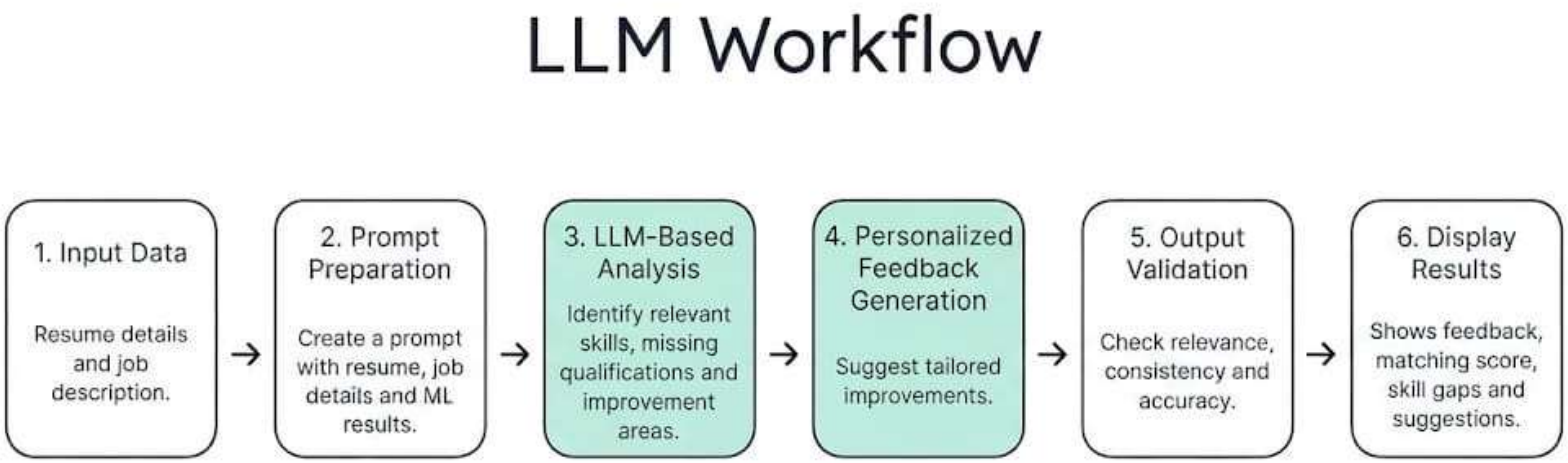


Fig. 3.5: NLP Workflow : Resume & Job Information Extraction

4. Advantages / Disadvantages

❖ Advantages

- Structured extraction of resume information.
- Support for scanned resumes through OCR.
- Comparison of the resume with job requirements.
- Identification of potential skill gaps.
- Personalized, evidence-based feedback.
- Reduced manual effort.

❖ Disadvantages

- OCR errors on poor-quality scans.
- NLP extraction errors; results depend on data quality.
- Limitations of the matching model.
- LLM may generate unsupported claims.
- Privacy and security of personal data.
- Human review is always needed.

5. Conclusion

The ML-Based ATS Resume Analyzer and Optimizer is a proposed system designed to help candidates improve their resumes and understand how well they match specific job requirements. The proposed workflow consists of four stages:
OCR extracts text from scanned or image-based resumes when required,
NLP processes and organizes resume and job information,
Machine Learning estimates compatibility and identifies potential skill gaps, and
LLM explains the results and suggests resume improvements.
These features are proposed and have not yet been experimentally evaluated. Therefore, no claims are made regarding system accuracy or superior performance compared with other models. The compatibility score is intended to provide guidance only and does not guarantee **ATS acceptance** or employment.

6. Project Initiation

Project Title: ML-Based Resume Optimizer

Project initiation defines the main problem, objectives, scope, stakeholders, and resources required for the proposed system.

The proposed project aims to develop a system that **analyzes resumes, compares candidate information** with **job requirements, identifies skill gaps,**and recommends **suitable job roles** and **resume improvements**.

The project planning phase helps identify the necessary technologies, tools, and resources required for implementation.

6. 1 Description of Problem Statement

❖ Problem

- Job seekers often cannot tell how well a resume matches a job description or which requirements are missing or weakly documented.
- Resumes come in many formats, including scanned images, which makes information extraction difficult.
- Manual evaluation is slow, and simple keyword matching can miss related wording.

❖ Problem Statement

To design a system that extracts information from resumes and job descriptions, estimates compatibility, identifies potential skill gaps, and gives evidence-based resume-improvement suggestions for user review.

6. 2 Scope and Objectives

❖ Scope of the Project

1. Accept resumes in supported document and image formats.
2. Extract text using OCR when required.
3. Identify skills, education, and work experience using NLP.
4. Compare resume information with job descriptions.
5. Identify skill gaps and suggest suitable job roles.
6. Provide recommendations for resume improvement.

❖ Objectives

1. Automate resume information extraction.
2. Analyze compatibility between resumes and job requirements.
3. Help candidates identify missing skills.
4. Support better resume preparation and job-role selection.

6. 3 Stake holder

The main stakeholders of the proposed ML-Based Resume Optimizer are :

1. **Job Seekers:** Upload resumes, review matching results, and improve their resumes.
2. **Recruiters and HR Teams:** Can use matching results to support initial resume screening.
3. **Project Developers:** Design, develop, test, and maintain the system.
4. **Educational Institutions:** Can use skill-gap information to guide students in developing relevant skills.
5. **System Administrator:** Manages system access, operation, and data security.

6. 4 Seminar Discussion Plan

The seminar discussion will cover the following topics :

1. Introduction to the ML-Based Resume Optimizer.
2. Literature survey and existing resume-screening approaches.
3. Overview of OCR, NLP, Machine Learning, and Recommendation Systems.
4. Proposed system architecture and working process.
5. Advantages, disadvantages, and applications.
6. Project scope, objectives, and required resources.
7. Conclusion and future improvements.

6.5 Platform/ Equipment/ Resources

Hardware Requirements

- Computer or laptop.
- Minimum 4 GB RAM; 8 GB or more recommended.
- Internet connection.

Software Requirements

- Operating System: Windows or Linux.
- Programming Language: Python.
- Development Environment: VS Code or Jupyter Notebook.
- NLP and ML Libraries: NLTK, spaCy, or scikit-learn.
- OCR Tool: Tesseract OCR or a suitable OCR library.

Other Resources

- Sample resumes for testing.
- Job descriptions for different roles.
- Resume–job matching data for evaluation.

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Thank you